GEODNET RTK Service Coordinate Systems

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GEODNET provides global and regional coordinate outputs through NTRIP mountpoints. This document records the coordinate frame and coordinate epoch configured by the service. It also distinguishes a frame transformation from a complete national datum realization.

The service outputs Earth-centered Earth-fixed or geodetic coordinates with ellipsoidal height. It does not by itself supply a country's projected mapping grid, vertical datum, geoid model, cadastral transformation, or legal survey procedure. Users doing control, cadastral, legal, or other high-accuracy work should validate the output against the responsible national authority.

For a comprehensive technical reference on national planar map projections (e.g., State Plane, UTM, Lambert-93, Gauss-Krüger) and national hybrid geoid models (e.g., GEOID18, OSGM15, GCG2016, AUSGeoid2020) used in engineering and surveying workflows, see Planar Coordinate Systems and Geoid Models Reference Guide.

1. Mountpoints

Mountpoint	Configured output
`AUTO`	Selects the regional output in Sections 2 and 3 from the station's three-character country or territory code, normally ISO 3166-1 alpha-3. Legacy aliases are also accepted.
`AUTO_ITRF2020`	ITRF2020 at the observation epoch.
`AUTO_ITRF2014`	ITRF2014 at the observation epoch.
`AUTO_WGS84`	Legacy WGS84(G2139) output at the midpoint of the observation year. See Section 4.
`BRDC`	Broadcast ephemeris data; this is not a coordinate-frame output.
`NATRF2022`	North America plate-fixed test service using NATRF2022. Confirm the coordinate epoch in service metadata.
`SIRGAS2000(YYYY.5)`	South America test service using SIRGAS2000 at the midpoint of the named year.

Mountpoint availability and coverage are operational service properties and may differ from the transformation catalogue below.

2. AUTO regional outputs

"Configured output" is the name emitted by the current converter when velocity is available. "Status" explains whether the computation is a direct frame-and-epoch transformation or has a known model or authority limitation.

#	Country or region	Code	Configured output	Basis	Status
1	United States - CONUS and Alaska	`USA`	`NAD83(2011)(2010.00)`	NAD83(2011), epoch 2010.0	NGS transformation; regional displacement behavior depends on HTDP and input velocity [R2].
2	United States - Hawaii	`USA`	`NAD83(PA11)(2010.00)`	NAD83(PA11), epoch 2010.0	NGS transformation; selected geographically within the USA route [R2].
3	Guam and Northern Mariana Islands	`GUM`, `MNP`	`NAD83(MA11)(2010.00)`	NAD83(MA11), epoch 2010.0	NGS transformation [R2].

#	Country or region	Code	Configured output	Basis	Status
4	American Samoa	ASM	NAD83(PA11)(2010.00)	NAD83(PA11), epoch 2010.0	NGS plate assignment [R2].
5	Puerto Rico and U.S. Virgin Islands	PRI, VIR	NAD83(2011)(2010.00)	NAD83(2011), epoch 2010.0	Production default during NSRS modernization. CATRF2022 remains a separate experimental transformation [R2, R31].
6	Canada	CAN	NAD83(CSRS)v8(2010.00)	NAD83(CSRS)v8, epoch 2010.0	EPSG operation 10415 at the common epoch; propagation uses station velocity, not NAD83v80VG [R4, R5].
7	Mexico	MEX	ITRF2008(2010.00)	ITRF2008, epoch 2010.0	Configured to the RGNA reference frame; validate against current INEGI products [R16].
8	Costa Rica	CRI, COS	CR-SIRGAS(2014.59)	ITRF2008/IGb08, epoch 2014.59	Official reference epoch; distinct from the operational ITRF2014 epoch 2019.24 [R7, R17].
9	Panama	PAN	SIRGAS-Panama(2011.60)	ITRF2008, epoch 2011.6	National realization listed by SIRGAS [R7].
10	Colombia	COL	MAGNA-SIRGAS(2018.00)	ITRF2014, epoch 2018.0	Current network basis documented by IGAC [R10].
11	Chile	CHL	SIRGAS-Chile(2021.00)	IGb14/ITRF2014, epoch 2021.0	National realization listed by SIRGAS; deformation validation remains necessary [R7].
12	Jamaica	JAM	JAD2001(2001.00)	Implemented as ITRF2000, epoch 2001.0	Exact centimetre-level identity has not been independently verified [R18].
13	Europe, except entries below	Listed European codes	ETRF2000(2010.00)	ETRF2000, epoch 2010.0	GEODNET service frame. Countries maintain distinct national ETRS89 realizations [R8].
14	France	FRA	ETRF2000(2019.00)	RGF93 v2b basis, epoch 2019.0	National realization basis [R9].
15	Italy	ITA	ETRF2000(2008.00)	RDN2008 basis, epoch 2008.0	National realization basis [R8].
16	Iceland	ISL	ISN2016-approx(2016.50)	IGS14, epoch 2016.5	Linear approximation; the Icelandic deformation treatment is not implemented [R6].
17	Turkey	TUR	ITRF96(2005.00)	TUREF/ITRF96, epoch 2005.0	Earthquake deformation requires authority treatment [R19].
18	Australia	AUS	GDA2020(2020.00)	ITRF2014, epoch 2020.0	Validate plate-motion handling for official work [R20].
19	New Zealand	NZL	NZGD2000-approx(2000.00)	ITRF96, epoch 2000.0	Linear approximation; LINZ deformation and earthquake patches are not implemented [R11].
20	Japan	JPN	JGD2011-approx(2011.39)	ITRF2008, epoch 2011-05-24	Approximation to JGD2011. GSI introduced JGD2024 on 2025-04-01 [R14].
21	China	CHN	CGCS2000(2000.00)	ITRF97, epoch 2000.0	Defining alignment; does not cover Hong Kong or Macao local systems [R21].
22	South Korea	KOR	KGD2002(2002.00)	ITRF2000, epoch 2002.0	Configured national-frame alignment [R22].
23	Taiwan	TWN	ITRF2020(2025.00)	ITRF2020, epoch 2025.0	GEODNET configuration. Mainland e-GNSS[2025] changed on 2025-05-02; exact network agreement and offshore-island coverage are not claimed [R23].

#	Country or region	Code	Configured output	Basis	Status
24	Thailand	THA	ITRF2014(2021.93)	ITRF2014, epoch 2021.93	CORS adjustment basis; not a claim about every cadastral product [R24].
25	Malaysia	MYS	GDM2020-approx(2020.00)	ITRF2014, epoch 2020.0	Linear approximation; JUPEM velocity and post-seismic procedures are not implemented [R13].
26	Indonesia	IDN	SRGI2013-approx(2012.00)	ITRF2008, epoch 2012.0	Linear approximation; BIG deformation model is not implemented [R12].
27	Philippines	PHL	PGD2020(2020.04)	ITRF2014, epoch 2020.044	PAGeNet port-specific realization; other ports may use different frames [R25].
28	India	IND	ITRF2008(2005.00)	ITRF2008, epoch 2005.0	GEODNET configuration; a current SOI specification for this exact epoch was not verified [R26].
29	Saudi Arabia	SAU	KSA-GRF17(2017.00)	ITRF2014, epoch 2017.0	National-frame alignment [R27].
30	United Arab Emirates	ARE	MTRF2000(2004.00)	ITRF2000, epoch 2004.0	Dubai-related configuration; not represented as the official frame of every emirate.
31	Georgia	GEO	ITRF2008(2011.353)	Georgia Geodetic Datum basis, epoch 2011.353	Compare velocity treatment with the national specification [R28].
32	Egypt	EGY	ITRF2008(2011.811)	ITRF2008, epoch 2011.811	Legacy CORS configuration; current ESA operational status was not independently verified.
33	Nigeria	NGA	NGD2012(2012.00)	Implemented as ITRF2008, epoch 2012.0	GEODNET configuration; nationwide legal adoption was not independently verified [R29].
34	South Africa	ZAF	ITRF1991(1994.00)	Hartebeesthoek94 basis, ITRF91 epoch 1994.0	Output name describes the implemented basis, not the national realization [R30].
35	Sri Lanka	LKA	GEODNET-SLCORS-fit(2024.00)	Empirical local fit	Four-station GEODNET fit; not an official WGS84 realization or SLD99 transformation.
36	Kenya	KEN	ITRF2014(observation epoch)	ITRF2014 at observation epoch	GEODNET configuration; current national CORS realization was not independently verified.
37	Hong Kong	HKG	HK-SatRef-approx(1998.32877)	ITRF96 at 1998:121 (May 1, 1998)	Linear frame-and-epoch approximation pending SatRef control validation; not HK1980 Grid or HKPD height [R32].
38	Macao	MAC	Macao-ITRF2005-approx(2008.37568)	ITRF2005, epoch 2008.37568	Linear approximation pending MoSRef control validation; not Macao Grid. Uses the authority's published decimal epoch [R33, R34].
39	Remaining three-character codes	Other	WGS84(G2139)(observation-year.50)	ITRF2014-equivalent legacy fallback	Global fallback, not the country's official datum. Missing or non-three-character codes are rejected.

3. Country-specific SIRGAS outputs

Country	Code	Configured output	Published basis
Argentina	ARG	POSGAR07(2006.60)	ITRF2005, epoch 2006.6 [R7]
Bolivia	BOL	MARGEN(1995.40)	SIRGAS95/ITRF94, epoch 1995.4 [R7]
Brazil	BRA	SIRGAS2000(2000.40)	ITRF2000, epoch 2000.4 [R7]
Colombia	COL	MAGNA-SIRGAS(2018.00)	ITRF2014, epoch 2018.0 [R7, R10]
Ecuador	ECU	RENAGE(1995.40)	SIRGAS95/ITRF94, epoch 1995.4 [R7]
El Salvador	SLV	SIRGAS-ES2007(2007.80)	ITRF2005/IGS05, epoch 2007.8 [R7]
Dominican Republic	DOM	ITRF2008(2016.40)	IGb08, epoch 2016.4 [R7]
Peru	PER	SIRGAS2000(2000.40)	ITRF2000, epoch 2000.4 [R7]
Uruguay	URY	SIRGAS-ROU98(1995.40)	SIRGAS95/ITRF94, epoch 1995.4 [R7]

Additional regional group assignments

These are service configurations, not verified national-datum assignments. The explicit country routes above take precedence.

- **ITRF2000 at epoch 2000.4:** ABW, BES, BLZ, CUW, FLK, GTM, GUF, GUY, HND, NIC, PRY, SUR, VEN.
- **NAD83(2011) at epoch 2010.0:** AIA, ATG, BRB, BMU, VGB, CYM, CUB, DMA, GRL, GRD, GLP, HTI, MTQ, MSR, ANT, BLM, KNA, LCA, MAF, SPM, VCT, SXM, BHS, TTO, TCA. ANT is a retained legacy code. French territories maintain separate official national references [R9].
- **ETRF2000 at epoch 2010.0:** AUT, BEL, BGR, HRV, CYP, CZE, DNK, EST, FIN, DEU, GRC, HUN, IRL, LVA, LTU, LUX, MLT, NLD, POL, PRT, ROU, SVK, SVN, ESP, SWE, ALB, AND, ARM, BLR, BIH, FRO, GIB, IMN, XKX, LIE, MKD, MDA, MCO, MNE, NOR, RUS, SMR, SRB, CHE, UKR, GBR, VAT. France and Italy use the epochs listed in Section 2.

4. Accuracy and interpretation

Outputs ending in -approx omit an authority deformation or earthquake model, or have not yet been validated against the local control network, as identified in the table. Linear propagation using a station velocity cannot reconstruct coseismic jumps or spatially varying deformation. Use the authority's model or transformation service when national-control agreement is required.

The converter applies published Helmert parameters between ITRF realizations [R1]. Regional propagation also depends on the availability and provenance of station velocity. Coordinate uncertainty reported by the upstream positioning solution does not automatically include transformation, velocity, or deformation-model uncertainty.

AUTO_WGS84 retains WGS84(G2139) for compatibility. NGA introduced WGS84(G2296), aligned with ITRF2020/IGS20, after G2139 [R15]. The service epoch is the midpoint of the observation year, which can differ from the current calendar year for reprocessed data.

The country code controls routing. A country code indicates jurisdiction, not necessarily the applicable legal datum for a particular project, agency, island, or service endpoint.

5. References

- **[R1] IERS/IGN**, ITRF2020 transformation parameters - Helmert parameters and rates.
- **[R2] NOAA NGS**, HTDP User Guide v3.6.0 - NAD83 transformations and displacement models.

- **[R3] NOAA NGS**, 2022 terrestrial reference frames - Euler parameters for NATRF2022, PATRF2022, CATRF2022 and MATRF2022.
- **[R4] NRCan**, Canadian reference-system updates - NAD83(CSRS)v8 and NAD83v80VG.
- **[R5] EPSG**, ITRF2020 to NAD83(CSRS)v8, operation 10415.
- **[R6] National Land Survey of Iceland**, ISNET2016 remeasurement report, printed page 36 - IGS14 at epoch 2016.5.
- **[R7] SIRGAS**, National networks and infrastructure.
- **[R8] BKG/EUREF**, National reference frames and geoid models - includes the EUREF overview, version 1.0c, 2025-06-30.
- **[R9] IGN France**, Official geodetic texts and frames.
- **[R10] IGAC Colombia**, MAGNA-SIRGAS network metadata.
- **[R11] LINZ**, NZGD2000 deformation model.
- **[R12] BIG Indonesia**, SRGI2013 deformation model.
- **[R13] JUPEM Malaysia**, GDM2020 propagation procedure.
- **[R14] GSI Japan**, JGD2024 adoption.
- **[R15] NGA**, WGS84(G2296) terrestrial reference update.
- **[R16] INEGI**, Red Geodesica Nacional Activa.
- **[R17] Registro Nacional Costa Rica**, CR-SIRGAS information - cites Executive Decree 40962-MJP.
- **[R18] National Land Agency Jamaica**, Digital cadastral surveying standards.
- **[R19] BKG CRS-EU**, TUREF record.
- **[R20] Geoscience Australia**, GDA2020.
- **[R21] Shenzhen Municipal Government**, CGCS2000 description.
- **[R22] NGII Korea**, Geodetic surveying overview.
- **[R23] NLSC Taiwan**, e-GNSS 2025 announcement - effective 2025-05-02; Penghu, Kinmen and Matsu retain the previous framework.
- **[R24] JICA**, Thailand CORS project report.
- **[R25] NAMRIA**, PAGeNet frame bulletin.
- **[R26] Survey of India**, CORS service.
- **[R27] GEOSA**, KSA-GRF17 FAQ.
- **[R28] NAPR Georgia**, Coordinate transformation guide.
- **[R29] OSGOF**, Geocentric datum transition paper.
- **[R30] South Africa NGI**, Geodesy and GPS.
- **[R31] NOAA NGS**, NSRS modernization transition and NAD83 frame assignments - NAD83(2011) for the Caribbean plate during the transition.
- **[R32] Hong Kong Lands Department**, SatRef coordinate list - ITRF96 at epoch 1998:121; HK1980 Grid requires the published local transformation.
- **[R33] Macao mapping authority**, Geodetic datum explanatory notes, Appendix 1 - ITRF2005 at published epoch 2008.37568. The legacy DSCC website is migrating to DSSCU.
- **[R34] Macao MoSRef / Hong Kong Lands Department**, MoSRef coordinate list, 2024-12-03 and Hong Kong-Macao transformation with a check example.

6. Feedback and change control

Corrections should include the responsible authority, realization name, coordinate epoch, source URL, publication date or version, and whether a deformation model is required. Send corrections to **Dr. Yudan Yi** at **yudan@geodnet.com**.

Transformation parameters and routing are maintained in the GEODNET coordinate converter. A documentation entry does not establish legal adoption or guaranteed accuracy; the cited authority remains definitive.